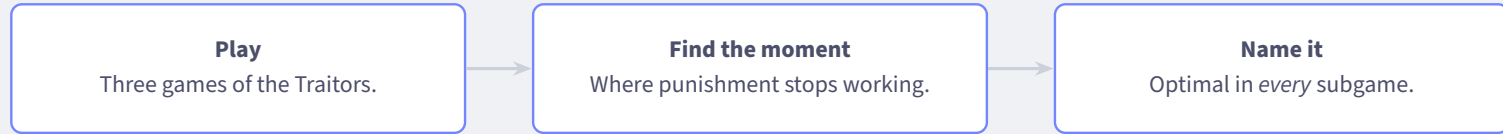


Subgame perfection

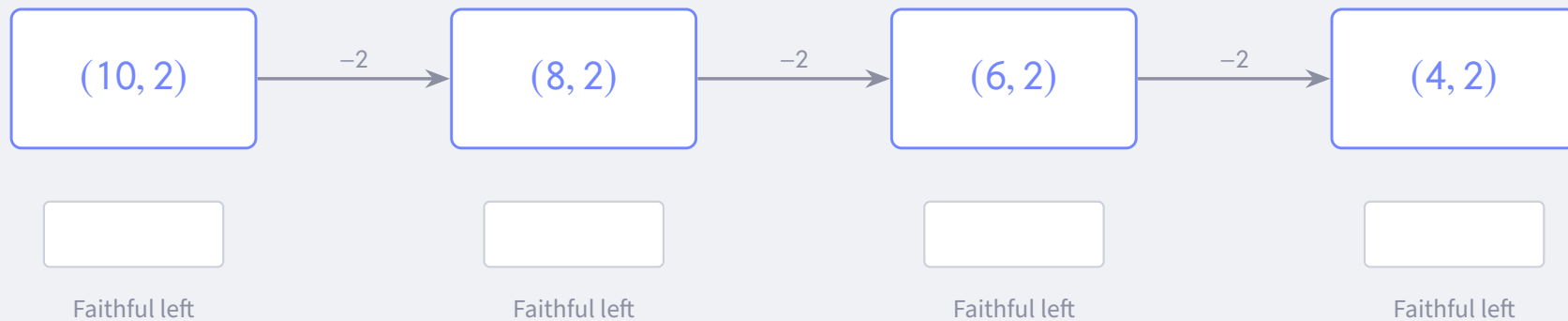
A threat that stops being worth carrying out



Players:

Traitors:

Who won?



Each round loses one player to the day vote and one to the night murder.

The Traitors win the moment $2m \geq n$, which first happens at . Reaching it takes rounds.

It makes deviation visible.

If a Traitor concentrates their vote, somebody gets votes and somebody gets . Everyone can see who broke the rule.

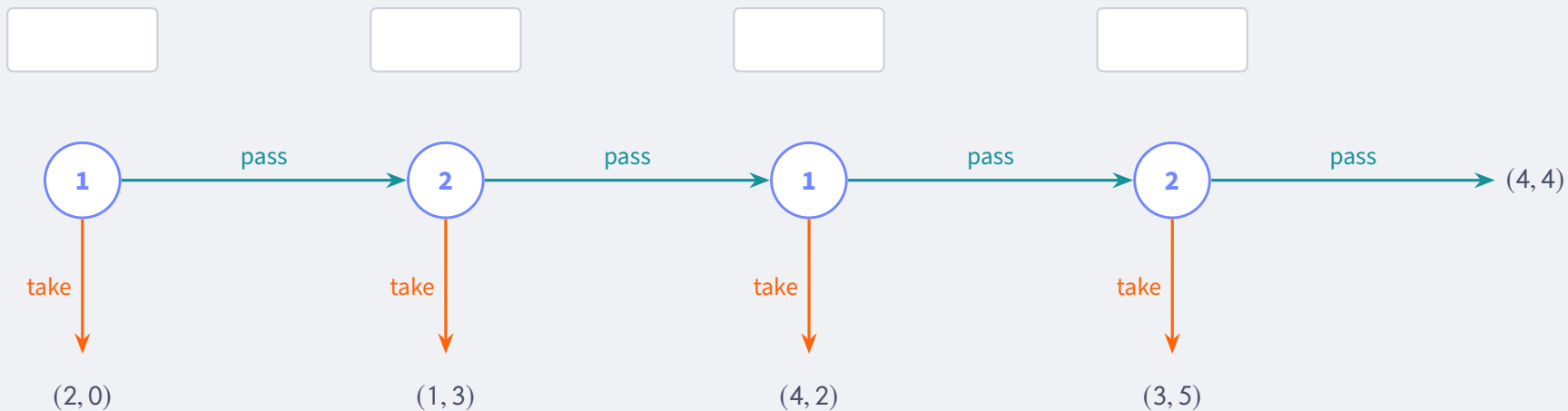
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At (6, 2) a Traitor who deviates is still detected. But after the night murder, are there enough Faithful left to banish them?

So the threat bites exactly while

$$n \quad \boxed{} \quad 2m + 2$$

Below that line the Traitors should stop hiding and collude in the open. That is their best response *in that subgame*, whether or not any game we played actually reached it.



Work right to left. Write the value of each node in the box above it.

Backwards induction gives: player 1 , player 2 , with payoffs .

Now take a different plan. Player 1 takes at both of their nodes. Player 2 takes at their first node, but *passes* at their last.

Is it a Nash equilibrium?

Is it subgame perfect?

Which node is never reached, and why does that let the bad promise survive?

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The Traitor complies or deviates; the Faithful then banish the suspect or carry on. The suspect is really the Traitor with probability q .

comply	$(1, 2)$
deviate, then carry on	$(3, 0)$
deviate, then banish	$(3 - 8q, 8q - 5)$

Draw the tree:

A rectangular box containing a grid of dots, intended for drawing a game tree. The grid consists of 10 rows and 20 columns of dots.

Banishing beats carrying on when $>$, that is q

The threat deters the Traitor when $<$, that is q

So (comply, banish) is a Nash equilibrium that is *not* subgame perfect exactly when q lies in .

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What we just did in the room is **Question 1** on the Subgame Perfection page.

- (a) Draw the extensive form. [3]
- (b) Backward induction: the subgame perfect equilibrium and its payoffs. [5]
- (c) Show (comply, banish) is a Nash equilibrium. [5]
- (d) Explain why it is not subgame perfect, and how that mirrors the class game. [6]
- (e) For general q , find where the threat deters and where it is credible. [4]

vknights.org/gt/topics/subgame-perfection.html